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信号与系统

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Complex Frequency Domain Analysis of Continuous Time Signals and Systems

It is known that the differential equation describing a continuous LTI system is $y''(t) + 2y'(t) - 3y(t) = x(t)$, input signal $x(t) = e^{-t}u(t)$, The initial state is $y(0^-)=4$, $y'(0^-)=2$, but:

※How to analyze the complete response of a continuous-time LTI system?

Time domain:
Zero input response (homogeneous solution)+ zero state response $[x(t)*h(t)]$

Frequency domain:
Only zero-state response, and requires the input signal to be absolutely integrable and the system stable

※How to analyze the system characteristics of a continuous-time LTI system?

Time domain:
Described by the impulse response h(t) of the system

Frequency domain:
Described by the frequency response H(jw) of the system
But only for stable systems.

Complex frequency domain analysis will effectively solve the above problems

Complex Frequency Domain Analysis of Continuous Time Signals and Systems

Complex Frequency Domain Analysis of Continuous Time Signals

- Complex frequency domain representation of continuous signals
- Commonly used one-sided Laplace transform
- Properties of one-sided Laplace transform
- One-sided inverse Laplace transform
- Bilateral Laplace transform and inverse transform

Complex Frequency Domain Analysis of Continuous LTI Systems

- Complex frequency domain description of continuous systems
- System functions and system characteristics
- Simulation of continuous time systems
- Complex frequency domain analysis of system response
- Realization of complex frequency domain analysis MATLAB

Complex Frequency Domain Representation of Continuous Time Signals

- Continuous-time signal Laplace transform
- Definition of one-sided Laplace transform
- The region of convergence of the one-sided Laplace transform

1. Laplace transform of continuous time signal

Complex frequency-domain representation of continuous-time signals: The signal $x(t)$ is represented as a linear combination of complex exponents e^{st}.

$$x(t) = \frac{1}{2\pi j}\int_{\sigma - j\infty}^{\sigma + j\infty} X(s)e^{st}ds \qquad s = \sigma + j\omega$$

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$$

$X(s)$ is a function of the complex frequency s, called the complex spectrum of the signal.

1. Laplace transform of continuous time signal

Laplace forward transform:

$$X(s)=\int_{-\infty}^{\infty}x(t)\mathrm{e}^{-st}\mathrm{d}t$$

Inverse Laplace transform:

$$x(t)=\frac{1}{2\pi\mathrm{j}}\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty}X(s)\mathrm{e}^{st}\mathrm{d}s$$

※Symbolic representation:

$$x(t)\xleftrightarrow{\mathscr{L}}X(s)$$

$$X(s)=\mathscr{L}[x(t)]$$

$$x(t)=\mathscr{L}^{-1}[X(s)]$$

Laplace,Pierre-Simon,1749-1827

French mathematician and astronomer. The family was poor, relying on the sponsorship of neighbors to go to school, became a member of the French Academy in 1816, and served as the president of the academy the following year. He mainly studies celestial mechanics and physics, and believes that mathematics is only a tool for solving problems, but creates and develops many new mathematical methods when using mathematics.

Main achievements: Expounded the movement of celestial bodies, the shape of the earth, planetary perturbation, the theory of moon departure and the three-body problem in "Celestial Mechanics", and introduced the famous Laplace equation. In "The Analytical Theory of Probability", he summarized the research on probability theory at that time, and discussed the application of probability in elections, trial investigations, meteorology and other aspects.

1. Continuous time signal Laplace transform

The relationship between Laplace transform and Fourier transform of continuous time signal

because $s=\sigma+j\omega$

$$X(s)=\int_{-\infty}^{\infty}x(t)\mathrm{e}^{-st}\mathrm{d}t \quad =\int_{-\infty}^{\infty}x(t)\mathrm{e}^{-(\sigma+\mathrm{j}\omega)t}\mathrm{d}t$$

$$=\int_{-\infty}^{\infty}x(t)\mathrm{e}^{-\sigma t}\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$=\mathscr{F}\{x(t)\mathrm{e}^{-\sigma t}\}$$

The Laplace transform is regarded as the Fourier transform of the signal $x(t)\,\mathrm{e}^{-\sigma t}$

1. Continuous time signal Laplace transform

The relationship between Laplace transform and Fourier transform of continuous time signal

For example $x(t) = \mathrm{e}^{at}u(t)$, $a > 0$ **Fourier transform?** does not exist!

Multiply $x(t)$ **by the attenuation factor** $\mathrm{e}^{-\sigma t}$ **and perform Fourier transform**

$$\mathscr{F}\{x(t)\mathrm{e}^{-\sigma t}\} = \int_{-\infty}^{\infty} x(t)\mathrm{e}^{-\sigma t}\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t \overset{s=\sigma+\mathrm{j}\omega}{=} \int_{0}^{\infty} \mathrm{e}^{\alpha t}\mathrm{e}^{-st}\mathrm{d}t$$

$$= \int_{0}^{\infty} \mathrm{e}^{-(s-\alpha)t}\mathrm{d}t = \frac{1}{s-\alpha}$$

As $\sigma > \alpha$

That is $$\mathscr{L}\{\mathrm{e}^{-\alpha t}u(t)\} = \frac{1}{s-\alpha}, \quad \sigma > \alpha$$

2. Definition of unilateral Laplace transform

In order to facilitate the description of the causal system and the complete response of the analysis system,

The following **unilateral Laplace transform** is often used:

$$X(s)=\int_{0^-}^{\infty} x(t)\mathrm{e}^{-st}\mathrm{d}t \qquad s=\sigma+\mathrm{j}\omega$$

$$x(t)=\frac{1}{2\pi\mathrm{j}}\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty} X(s)\mathrm{e}^{st}\mathrm{d}s$$

3. Convergence region of unilateral Laplace transform

$$\boxed{X(s)=\int_{0^-}^{\infty} x(t)\mathrm{e}^{-st}\mathrm{d}t}$$

■ If the unilateral Laplace transform of $x(t)$exists, the above integral needs to converge. Therefore, **the necessary and sufficient conditions** for the existence of unilateral Laplace transform are:

$$\int_{0^-}^{\infty}|x(t)\mathrm{e}^{-st}|\mathrm{d}t=\int_{0^-}^{\infty}|x(t)\mathrm{e}^{-\sigma t}|\mathrm{d}t=C,\qquad \sigma=\mathrm{Re}(s)$$

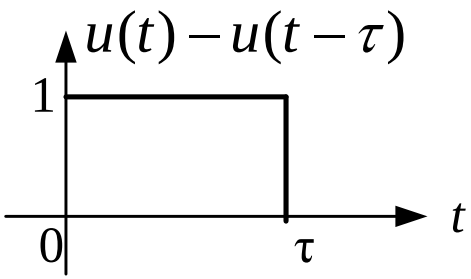
■The value range of σ where the above formula holds is called the convergence region of Laplace transform, abbreviated as ROC (Region Of Convergence).

3. Convergence region of unilateral Laplace transform

Example: Try to find the unilateral Laplace transform and its convergence region of the continuous signal *u*(*t*)-*u*(*t*-*t*).

Solution:

$$X(s)=\int_{0^-}^{\infty}[u(t)-u(t-\tau)]\mathrm{e}^{-st}\mathrm{d}t$$



$$=\int_{0^-}^{\tau}\mathrm{e}^{-st}\mathrm{d}t=-\frac{1}{s}\mathrm{e}^{-st}\Big|_{0^-}^{\tau}$$

$$=\frac{1}{s}(1-\mathrm{e}^{-s\tau})$$

$u(t)-u(t-\tau)$ *The region of convergence is the s-full plane, that is:* $\mathrm{Re}(s)>-\infty$

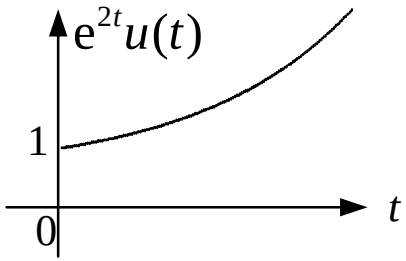
3. Convergence region of unilateral Laplace transform

Example: try to solve the unilateral Laplace transform of signal $e^{2t}u(t)$ and its convergence region.

Solution: $X(s)=\int_{0-}^{\infty} e^{2t}u(t)e^{-st}dt$

$$=\frac{1}{2-s}e^{(2-s)t}\Big|_{0^-}^{\infty}$$

$$=\frac{1}{s-2} \qquad \mathrm{Re}(s)>2$$

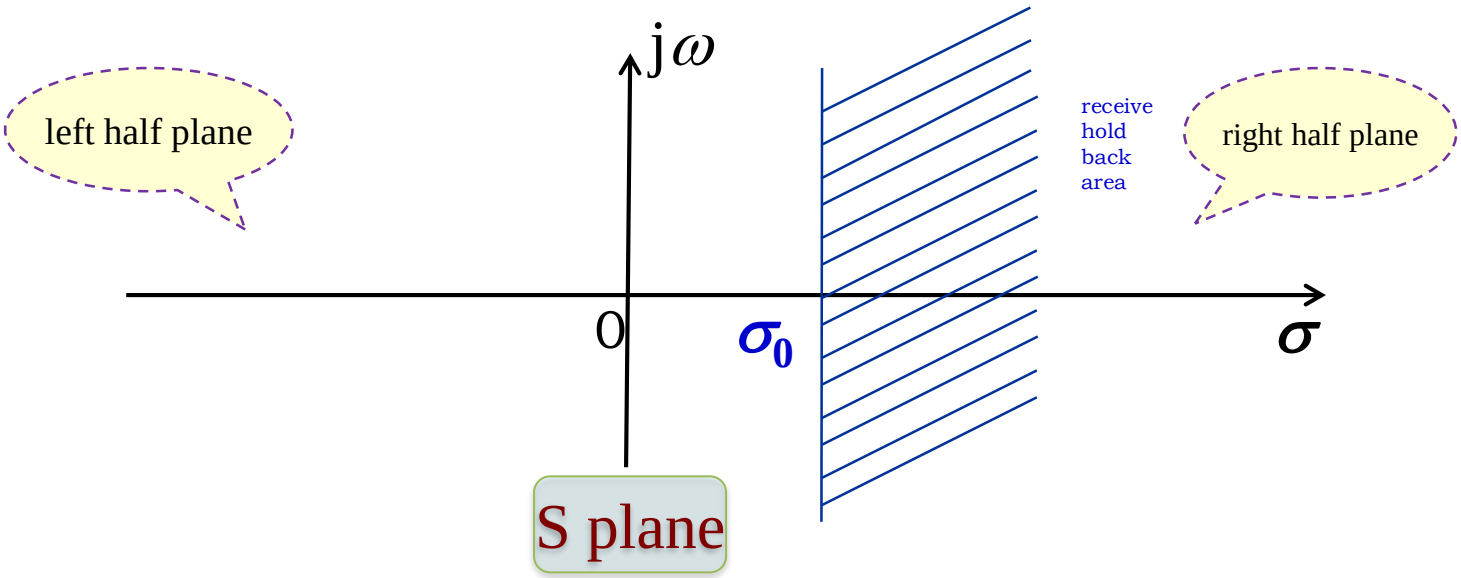


The Laplace transform of $e^{2t}u(t)$ exists , if and only if $\mathrm{Re}(s)>2$

3. Convergence region of unilateral Laplace transform

☼The region of convergence of the unilateral Laplace transform is expressed as Re(s) > σ_0 (σ_0 is a real constant).

☼The Region of Convergence (ROC) can be visualized using the S-complex plane.



One-sided Laplace transform of common signals

- Impulse signal
- Step signal
- Index signal
- Sinusoidal signal
- Ramp signal

1. Impulse signal

※Impulse signal $\delta(t), \quad \delta^{(n)}(t)$

$$\mathscr{L}[\delta(t)] = \int_{0^-}^{\infty} \delta(t) \mathrm{e}^{-st} \mathrm{d}t = \mathrm{e}^{-st}\Big|_{t=0} = 1 \qquad \mathrm{Re}(s) > -\infty$$

$$\delta(t) \xleftrightarrow{\mathscr{L}} 1, \qquad \mathrm{Re}(s) > -\infty$$

In the same way:

$$\delta^{(n)}(t) \xleftrightarrow{\mathscr{L}} s^n, \qquad \mathrm{Re}(s) > -\infty$$

2. Step signal

※ Step signal $u(t)$

$$\mathscr{L}[u(t)] = \int_{0^-}^{\infty} u(t)\mathrm{e}^{-st}\mathrm{d}t = \int_{0^-}^{\infty} \mathrm{e}^{-st}\mathrm{d}t$$

$$= -\frac{1}{s}\mathrm{e}^{-st}\Big|_{0^-}^{\infty} = \frac{1}{s} \qquad \mathrm{Re(s)} > 0$$

$$u(t) \xleftrightarrow{\mathscr{L}} \frac{1}{s}, \qquad \mathrm{Re}(s) > 0$$

3. Exponential signals

※ Index signal $e^{\lambda t}u(t)$， λcan be any plural

$$\mathscr{L}[e^{\lambda t}u(t)] = \int_{0^-}^{\infty} e^{\lambda t} e^{-st} dt$$

$$= -\frac{1}{s-\lambda} e^{(\lambda - s)t} \Big|_{0^-}^{\infty} = \frac{1}{s-\lambda} \qquad \mathrm{Re}(s) > \mathrm{Re}(\lambda)$$

$$e^{\lambda t}u(t) \xleftrightarrow{\mathscr{L}} \frac{1}{s-\lambda}, \quad \mathrm{Re}(s) > \mathrm{Re}(\lambda)$$

3. Exponential signals

※ Index signal $e^{\lambda t}u(t)$, λ can be any plural

$$e^{\lambda t}u(t) \xleftrightarrow{\mathscr{L}} \frac{1}{s-\lambda}, \quad \mathrm{Re}(s) > \mathrm{Re}(\lambda)$$

when $\lambda=\alpha$, Real Exponential Signal

$$e^{\alpha t}u(t) \xleftrightarrow{\mathscr{L}} \frac{1}{s-\alpha} \qquad \mathrm{Re}(s) > \alpha$$

when $\lambda=\mathrm{j}\omega_0$, Imaginary index signal

$$e^{\mathrm{j}\omega_0 t}u(t) \xleftrightarrow{\mathscr{L}} \frac{1}{s-\mathrm{j}\omega_0} \qquad \mathrm{Re}(s) > 0$$

when $\lambda=\alpha+\mathrm{j}\omega_0$, complex exponential signal

$$e^{(\alpha+\mathrm{j}\omega_0)t}u(t) \xleftrightarrow{\mathscr{L}} \frac{1}{s-(\alpha+\mathrm{j}\omega_0)} \qquad \mathrm{Re}(s) > \alpha$$

4. Sine signal

※ sine signal $\cos(\omega_0 t)u(t)$, $\sin(\omega_0 t)u(t)$

$$\cos(\omega_0 t)\,u(t) = \frac{\mathrm{e}^{\mathrm{j}\omega_0 t} + \mathrm{e}^{-\mathrm{j}\omega_0 t}}{2}u(t)$$

$$\xleftrightarrow{\mathscr{L}} \quad \frac{1}{2}\left(\frac{1}{s-\mathrm{j}\omega_0} + \frac{1}{s+\mathrm{j}\omega_0}\right) = \frac{s}{s^2+\omega_0^{\,2}}, \quad \mathrm{Re}(s) > 0$$

$$\sin(\omega_0 t)\,u(t) = \frac{\mathrm{e}^{\mathrm{j}\omega_0 t} - \mathrm{e}^{-\mathrm{j}\omega_0 t}}{2\mathrm{j}}u(t)$$

$$\xleftrightarrow{\mathscr{L}} \quad \frac{1}{2\mathrm{j}}\left(\frac{1}{s-\mathrm{j}\omega_0} - \frac{1}{s+\mathrm{j}\omega_0}\right) = \frac{\omega_0}{s^2+\omega_0^{\,2}}, \quad \mathrm{Re}(s) > 0$$

5. Ramp signal

※ Ramp signal $tu(t)$

$$\mathscr{L}[tu(t)] = \int_{0^-}^{\infty} t\mathrm{e}^{-st}\mathrm{d}t$$

$$= -\frac{t}{s}(\mathrm{e}^{-st})\Big|_{0^-}^{\infty} + \frac{1}{s}\int_{0^-}^{\infty} \mathrm{e}^{-st}\mathrm{d}t$$

$$= \frac{1}{s}\int_{0^-}^{\infty} \mathrm{e}^{-st}\mathrm{d}t = -\frac{1}{s^2}\mathrm{e}^{-st}\Big|_{0^-}^{\infty} = \frac{1}{s^2} \qquad \mathrm{Re}(s) > 0$$

$$tu(t) \xleftrightarrow{\mathscr{L}} \frac{1}{s^2}, \quad \mathrm{Re}(s) > 0$$

Properties of one-sided Laplace transform

- Linearity
- Time-shifting characteristics
- Shrinkage characteristics
- Convolution characteristics
- Product property
- Exponential Weighting Properties
- Linear weighting feature
- Differential characteristics
- Integral characteristics
- Initial value and final value theorem

1. Linear characteristics

► Linearity

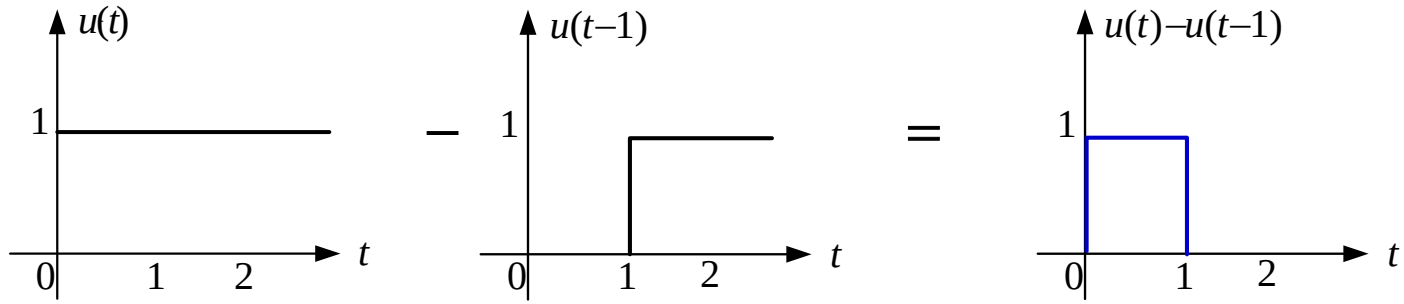
As $$x_1(t) \xleftrightarrow{\mathscr{L}} X_1(s), \quad \mathrm{Re}(s) > \sigma_1$$

$$x_2(t) \xleftrightarrow{\mathscr{L}} X_2(s), \quad \mathrm{Re}(s) > \sigma_2$$

then $$\alpha x_1(t) + \beta x_2(t) \xleftrightarrow{\mathscr{L}} \alpha X_1(s) + \beta X_2(s)$$

$$\mathrm{Re}(s) > \max(\sigma_1, \sigma_2)$$

Example: Try to solve the one-sided Laplace transform of $u(t)$-$u(t$-1) by using the linear property.



As $\quad u(t) \xleftrightarrow{\mathscr{L}} \dfrac{1}{s}, \ \mathrm{Re}(s) > 0$

And $\quad u(t-1) \xleftrightarrow{\mathscr{L}} \dfrac{1}{s}\mathrm{e}^{-s}, \quad \mathrm{Re}(s) > 0$

Using the linear property of the Laplace transform, we can get

$$u(t) - u(t-1) \xleftrightarrow{\mathscr{L}} \frac{1}{s} - \frac{1}{s}\mathrm{e}^{-s} = \frac{1-\mathrm{e}^{-s}}{s}, \quad \mathrm{Re}(s) > -\infty$$

2. Time-shifting property

► Time-shifting properties

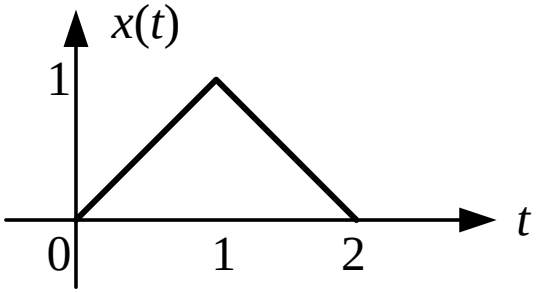
As $x(t) \xleftrightarrow{\mathscr{L}} X(s)$, $\mathrm{Re}(s) > \sigma_0$

then $x(t-t_0)u(t-t_0) \xleftrightarrow{\mathscr{L}} \mathrm{e}^{-st_0} X(s)$ $(t_0 > 0)$

$\mathrm{Re}(s) > \sigma_0$

Example: Solve the one-sided Laplace transform of $x(t)$ by using time-shift and linear characteristics.

Solution: $x(t)$ can be expressed by the basic signal as



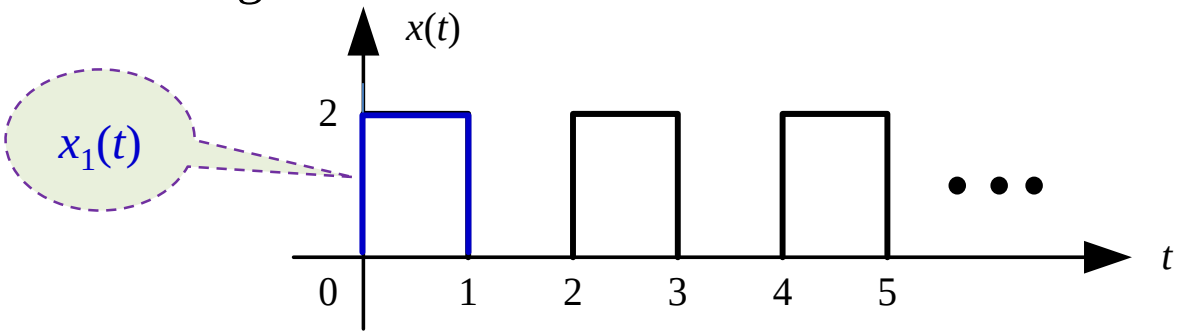
$$x(t) = r(t) - 2r(t-1) + r(t-2)$$

Known $\quad r(t) = tu(t) \xleftrightarrow{\mathscr{L}} \dfrac{1}{s^2}, \quad \mathrm{Re}(s) > 0$

Using the Time-shift and Linear characteristics of Laplace transform, we can get:

$$X(s) = \frac{1}{s^2}(1 - 2\mathrm{e}^{-s} + \mathrm{e}^{-2s}) = \frac{1 - 2\mathrm{e}^{-s} + \mathrm{e}^{-2s}}{s^2} \qquad \mathrm{Re}(s) > -\infty$$

Example: Try to find the one-sided Laplace transform of the signal shown in the figure.



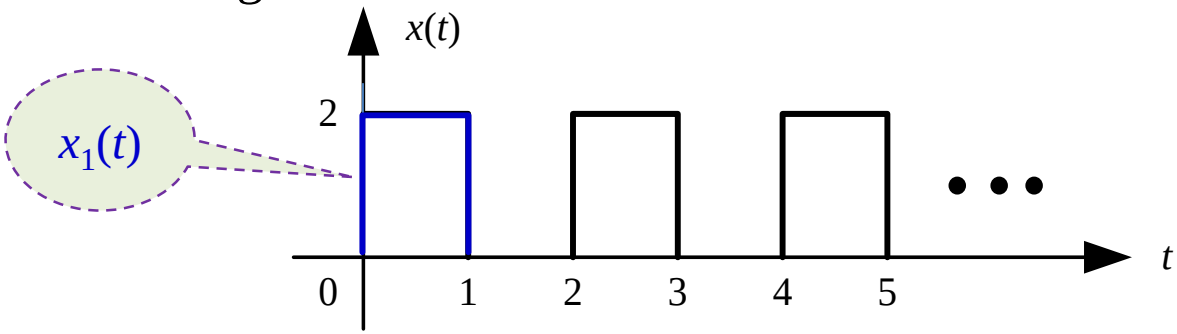
Analysis: A unilateral periodic signal $x_1(t)$ with period T can be expressed as a linear combination of the first periodic signal $x_1(t)$ and its time shift $x_1(t-kT)$, namely

$$x(t)=\sum_{n=0}^{\infty}x_1(t-nT)$$

If the Laplace transform $X_1(s)$ of $x_1(t)$ is calculated, using the time-shift and linear characteristics of Laplace transform, the Laplace transform of the given signal can be obtained as

$$\mathscr{L}[x(t)]=\sum_{n=0}^{\infty}X_1(s)\cdot \mathrm{e}^{-snT}=\frac{X_1(s)}{1-\mathrm{e}^{-sT}} \qquad \mathrm{Re}(s)>0$$

Example: Try to find the one-sided Laplace transform of the signal shown in the figure.



Solution: $x_1(t) = 2[u(t) - u(t-1)]$

Hence $$x_1(t) \xleftrightarrow{\mathscr{L}} \frac{2}{s}(1-\mathrm{e}^{-s}), \qquad \mathrm{Re}(s) > -\infty$$

Due to $$x(t) = x_1(t) + x_1(t-2) + x_1(t-4) + \cdots$$

So $$X(s) = X_1(s)(1+\mathrm{e}^{-2s}+\mathrm{e}^{-4s}+\cdots) = \frac{X_1(s)}{1-\mathrm{e}^{-2s}} \qquad \mathrm{Re}(s) > 0$$

3. Scaling property

► Scaling properties

As $$x(t) \xleftrightarrow{\mathscr{L}} X(s) \quad \mathrm{Re}(s) > \sigma_0$$

Then $$x(at) \xleftrightarrow{\mathscr{L}} \frac{1}{a} X(\frac{s}{a}), \qquad (a > 0)$$

$$\mathrm{Re}(s) > a\sigma_0$$

4. Convolution property

- Convolution properties

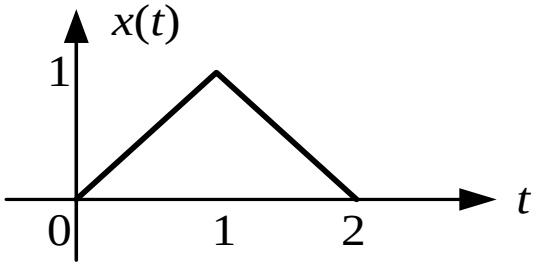
If $x_1(t) \xleftrightarrow{\mathscr{L}} X_1(s), \quad \mathrm{Re}(s) > \sigma_1$

$x_2(t) \xleftrightarrow{\mathscr{L}} X_2(s), \quad \mathrm{Re}(s) > \sigma_2$

then $x_1(t) * x_2(t) \xleftrightarrow{\mathscr{L}} X_1(s) X_2(s)$

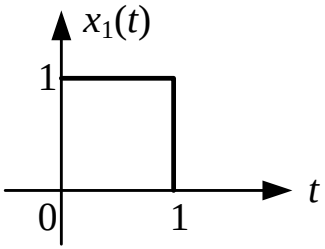
$\mathrm{Re}(s) > \max(\sigma_1, \sigma_2)$

Example: Try to use the convolution feature to solve the one-sided Laplace transform of the signal shown in the figure:



Solution:

$x(t)$ can be expressed as $x(t) = x_1(t) * x_1(t)$



In, $x_1(t) = u(t) - u(t-1)$

$$x_1(t) \xleftrightarrow{\mathscr{L}} X_1(s) = \frac{1-e^{-s}}{s}, \quad \mathrm{Re}(s) > -\infty$$

Therefore, using the convolution characteristics of the Laplace transform, we can get

$$X(s) = X_1(s) \cdot X_1(s) = \left(\frac{1-e^{-s}}{s}\right)^2, \quad \mathrm{Re}(s) > -\infty$$

5. Product property

- Product property

If $$x_1(t) \xleftrightarrow{\mathscr{L}} X_1(s), \quad \mathrm{Re}(s) > \sigma_1$$

$$x_2(t) \xleftrightarrow{\mathscr{L}} X_2(s), \quad \mathrm{Re}(s) > \sigma_2$$

then $$x_1(t)x_2(t) \xleftrightarrow{\mathscr{L}} \frac{1}{2\pi \mathrm{j}}[X_1(s) * X_2(s)]$$

$$\mathrm{Re}(s) > \sigma_1 + \sigma_2$$

6. Exponential weighting property

► Exponential Weighting Properties

If $x(t) \xleftrightarrow{\mathscr{L}} X(s), \quad \mathrm{Re}(s) > \sigma_0$

Then $\mathrm{e}^{-\lambda t} x(t) \xleftrightarrow{\mathscr{L}} X(s+\lambda)$

$$\mathrm{Re}(s) > \sigma_0 - \mathrm{Re}(\lambda)$$

Example: Use the exponential weighting property to solve the one-sided Laplace transform of the following signal: $e^{-at}\cos(\omega_0 t)u(t)$, where a is a real number.

Solution: The Laplace transform of the one-sided cosine signal is

$$\cos(\omega_0 t)u(t) \xleftrightarrow{\mathscr{L}} \frac{s}{s^2+\omega_0{}^2}, \quad \mathrm{Re}(s)>0$$

According to the exponential weighting characteristics of Laplace transform, we can get:

$$\mathscr{L}\{e^{-\alpha t}\cos(\omega_0 t)u(t)\} = \frac{s+\alpha}{(s+\alpha)^2+\omega_0{}^2},$$

$$\mathrm{Re}(s) > -\alpha$$

7. Linear weighting property

►linear weighting feature

if $\quad x(t) \xleftrightarrow{\mathscr{L}} X(s), \quad \mathrm{Re}(s) > \sigma_0$

then $\quad tx(t) \xleftrightarrow{\mathscr{L}} -\dfrac{\mathrm{d}X(s)}{\mathrm{d}s}, \quad \mathrm{Re}(s) > \sigma_0$

Example: Solve the unilateral Laplace transform of the following signals using the linear weighting property: $tu(t)$, $t^n u(t)$, $te^{-\lambda t}u(t)$, $t^n e^{-\lambda t}u(t)$, n is a positive integer.

Solution:

$$u(t) \xleftrightarrow{\mathscr{L}} \frac{1}{s}, \quad \mathrm{Re}(s) > 0$$

According to the linear weighting characteristics of unilateral Laplace transform, we can get:

$$\mathscr{L}\{tu(t)\} = -\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{s}\right) = \frac{1}{s^2}, \quad \mathrm{Re}(s) > 0$$

Repeatedly using the linear weighting feature, we can conclude:

$$\mathscr{L}\{t^n u(t)\} = \mathscr{L}\{t \cdot t^{n-1} u(t)\} = \frac{n!}{s^{n+1}}, \quad \mathrm{Re}(s) > 0$$

Example: Solve the unilateral Laplace transform of the following signals using the linear weighting property: $tu(t)$, $t^n u(t)$, $te^{-\lambda t}u(t)$, $t^n e^{-\lambda t}u(t)$, n is a positive integer.

Solution:

$$e^{-\lambda t}u(t) \xleftrightarrow{\mathscr{L}} \frac{1}{s+\lambda}, \quad \mathrm{Re}(s) > -\mathrm{Re}(\lambda)$$

According to the linear weighting property of unilateral Laplace transform, we can get:

$$\mathscr{L}(te^{-\lambda t}u(t)\} = -\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{s+\lambda}\right) = \frac{1}{(s+\lambda)^2}, \quad \mathrm{Re}(s) > -\mathrm{Re}(\lambda)$$

Repeatedly using the linear weighting property, we can get:

$$\mathscr{L}\{t^n e^{-\lambda t}u(t)\} = \frac{n!}{(s+\lambda)^{n+1}}, \quad \mathrm{Re}(s) > -\mathrm{Re}(\lambda)$$

8. Differential characteristics

► Differential property

If $\quad x(t) \xleftrightarrow{\mathscr{L}} X(s), \quad \mathrm{Re}(s) > \sigma_0$

then $\dfrac{\mathrm{d}x(t)}{\mathrm{d}t} \xleftrightarrow{\mathscr{L}} sX(s) - x(0^-), \quad \mathrm{Re}(s) > \sigma_0$

Prove: $\mathscr{L}\left[\dfrac{\mathrm{d}x(t)}{\mathrm{d}t}\right] = \int_{0^-}^{\infty} \dfrac{\mathrm{d}x(t)}{\mathrm{d}t} \mathrm{e}^{-st} \mathrm{d}t$

$$= x(t)\mathrm{e}^{-st}\Big|_{0^-}^{\infty} - \int_{0^-}^{\infty} x(t)(-s\mathrm{e}^{-st})\mathrm{d}t$$

$$= -x(0^-) + s\int_{0^-}^{\infty} x(t)\mathrm{e}^{-st}\mathrm{d}t = sX(s) - x(0^-) \qquad \mathrm{Re}(s) > \sigma_0$$

8. Differential property

►Differential property

Repeatedly applying the differential property, we get:

$$\frac{\mathrm{d}^2 x(t)}{\mathrm{d}t^2} \xleftrightarrow{\mathscr{L}} s^2 X(s) - sx(0^-) - x'(0^-)$$

$$\frac{\mathrm{d}^n x(t)}{\mathrm{d}t^n} \xleftrightarrow{\mathscr{L}} s^n X(s) - s^{n-1} x(0^-) - \cdots - sx^{(n-2)}(0^-) - x^{(n-1)}(0^-)$$

The relationship between ramp signal $tu(t)$, step signal $u(t)$, impulse signal $\delta(t)$ and its n order derivative $\delta^{(n)}(t)$ in s domain

ramp signal $tu(t)$

$\mathscr{L}$

$$\frac{1}{s^2},\ \mathrm{Re}(s) > 0$$

differential

step signal $u(t)$

$\mathscr{L}$

$$\frac{1}{s},\ \mathrm{Re}(s) > 0$$

differential

impulse signal $\delta(t)$

$\mathscr{L}$

$$1,\ \mathrm{Re}(s) > -\infty$$

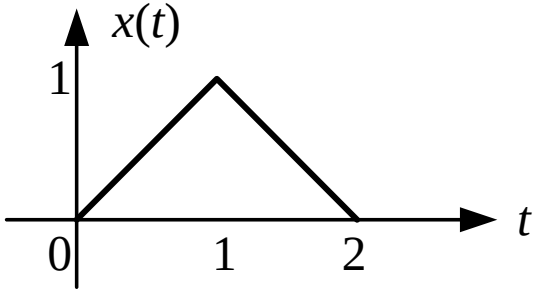
differential

n order of impulse function Derivative, $\delta^{(n)}(t)$

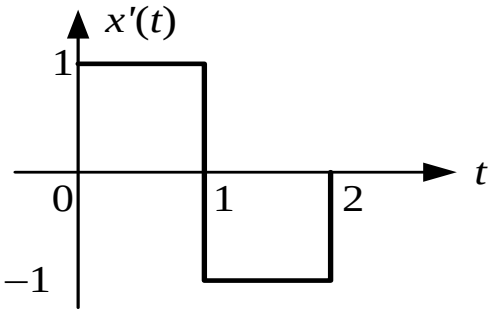
$\mathscr{L}$

$$s^n,\ \mathrm{Re}(s) > -\infty$$

Example: Use differential property to solve the one-sided Laplace transform of the signal shown in the figure:

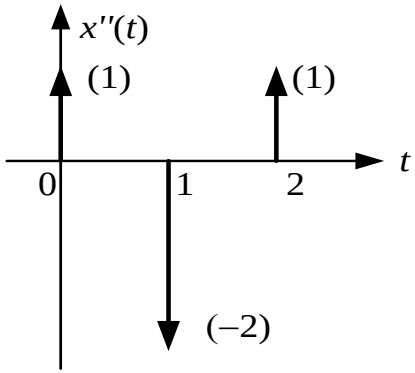


Solution: According to the differential characteristics of unilateral Laplace transform:



$$\frac{\mathrm{d}^2 x(t)}{\mathrm{d}t^2} \xleftrightarrow{\mathscr{L}} s^2 X(s) - s x(0^-) - x'(0^-)$$

$$= s^2 X(s) = 1 - 2\mathrm{e}^{-s} + \mathrm{e}^{-2s}$$



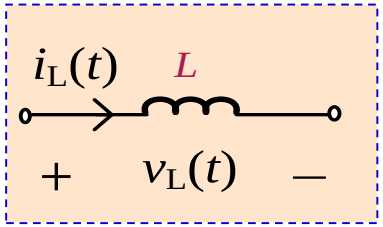
Therefore,

$$X(s) = \frac{1 - 2\mathrm{e}^{-s} + \mathrm{e}^{-2s}}{s^2}$$

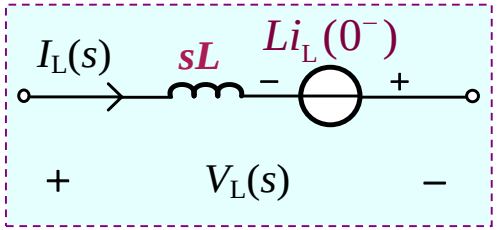
$$\mathrm{Re}(s) > -\infty$$

Application of Differential Property in Practical Circuit System

Inductive device



s-domain model of inductor



$$v_L(t) = L\frac{\mathrm{d}i_L(t)}{\mathrm{d}t} \longrightarrow V_L(s) = sLI_L(s) - Li_L(0^-)$$

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} \xleftrightarrow{\mathscr{L}} sX(s) - x(0^-)$$

9. Integral property

► Integral characteristics

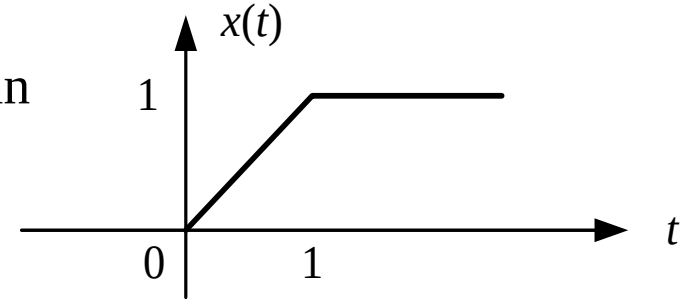
If $x(t) \xleftrightarrow{\mathscr{L}} X(s), \quad \mathrm{Re}(s) > \sigma_0$

Then $\int_{-\infty}^{t} x(\tau)\mathrm{d}\tau \xleftrightarrow{\mathscr{L}} \frac{X(s)}{s} + \frac{x^{(-1)}(0^-)}{s}$

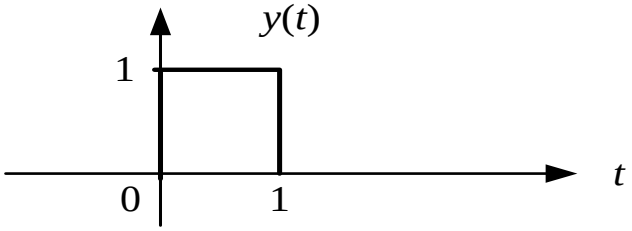
$\mathrm{Re}(s) > \max(\sigma_0, 0)$

If $x^{(-1)}(0^-)=0$, then $\int_{-\infty}^{t} x(\tau)\mathrm{d}\tau \xleftrightarrow{\mathscr{L}} \frac{X(s)}{s}$

Example: Find the signal $x(t)$ as shown in the figure one-sided Laplace transform.



Solution: $x(t) = \int_{-\infty}^{t} y(t)\mathrm{d}t$



because

$$y(t) = u(t) - u(t-1) \xleftrightarrow{\mathscr{L}} \frac{1-\mathrm{e}^{-s}}{s}, \quad \mathrm{Re}(s) > -\infty$$

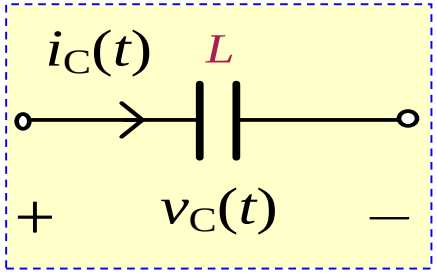
and $$x^{(-1)}(0^-) = \int_{-\infty}^{0^-} y(t)\mathrm{d}t = 0$$

According to the integral property of the Laplace transform:

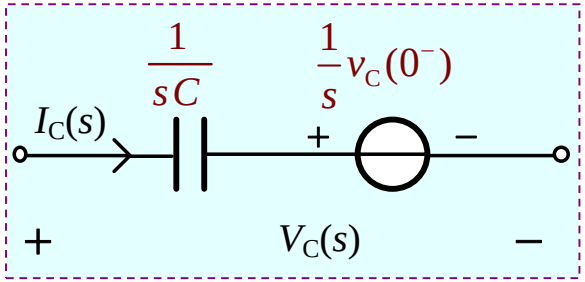
$$X(s) = \frac{Y(s)}{s} = \frac{1-\mathrm{e}^{-s}}{s^2}, \quad \mathrm{Re}(s) > 0$$

Application of Integral Property in Practical Circuit System

Capacitance



s-domain model of capacitance



$$v_C(t) = \frac{1}{C}\int_{-\infty}^{t} i_C(\tau)\mathrm{d}\tau \longrightarrow V_C(s) = \frac{1}{sC}I_C(s) + \frac{1}{s}v_C(0^-)$$

$$\int_{-\infty}^{t} x(\tau)\mathrm{d}\tau \xleftrightarrow{\mathscr{L}} \frac{X(s)}{s} + \frac{x^{(-1)}(0^-)}{s}$$

10. Initial value theorem and final value theorem

► Initial value and final value theorem

$$x(t) \xleftrightarrow{\mathscr{L}} X(s), \quad \mathrm{Re}(s) > \sigma_0$$

If $x(t)$ does not contain impulse and its derivatives at time $t = 0$, then

$$\lim_{t \to 0} x(t) = x(0^+) = \lim_{s \to +\infty} sX(s)$$

If the region of convergence of $sX(s)$ includes the $\mathrm{j}\omega$ axis, then

$$\lim_{t \to +\infty} x(t) = x(+\infty) = \lim_{s \to 0} sX(s)$$

Example: Given , $X(s)=\dfrac{1}{s+1}$, $\mathrm{Re}(s)>-1$

find the initial and final values of $x(t)$.

Solution: According to the initial value theorem, we can get

$$x(0^+)=\lim_{s\to+\infty} sX(s)=1$$

The convergence region of $sX(s)$ includes the jω axis, which can be obtained by directly applying the final value theorem

$$x(+\infty)=\lim_{s\to 0} sX(s)=0$$

Example: Given , $X(s)=\dfrac{1}{s+1}$, $\mathrm{Re}(s)>-1$

find the initial and final values of $x(t)$.

Solution: Since $X(s)$ is not a true fraction, $x(t)$ contains an impulse signal at $t=0$, The initial value theorem cannot be directly applied

Rewrite $X(s)$ as

$$X(s)=\frac{s}{s+1}=1-\underbrace{\frac{1}{s+1}}_{X_1(s)}$$

Applying the initial value theorem to $X_1(s)$ gives

$$x(0^+)=\lim_{s\to+\infty} sX_1(s)=1$$

The convergence region of $sX(s)$ includes the jω axis, which can be obtained by directly applying the final value theorem

$$x(+\infty)=\lim_{s\to 0} sX(s)=0$$

One-sided inverse Laplace transform

$$x(t)=\frac{1}{2\pi \mathrm{j}}\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty}X(s)\mathrm{e}^{st}\mathrm{d}s$$

※ **Residue method**

The residue method is more complicated to calculate, but has a wider scope of application.

※ **Partial fraction expansion**

The partial fraction method is relatively simple to solve, but it is generally only applicable to rational fractions.

1. Residue method

※Residue method

$$x(t)=\begin{cases}\dfrac{1}{2\pi \mathrm{j}}\displaystyle\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty} X(s)\mathrm{e}^{st}\mathrm{d}s=\sum_{k=0}^{n}\operatorname*{Res}_{s=p_k}[X(s)\mathrm{e}^{st}] & t>0\\ 0 & t<0\end{cases}$$

If p_k is a $X(s)\mathrm{e}^{st}$ single pole of , then:

$$\operatorname*{Res}_{s=p_k}[X(s)\mathrm{e}^{st}]=[(s-p_k)X(s)\mathrm{e}^{st}]_{s=p_k}$$

If p_k is the $X(s)\mathrm{e}^{st}$ m-fold pole of , then:

$$\operatorname*{Res}_{s=p_k}[X(s)\mathrm{e}^{st}]=\frac{1}{(m-1)!}\frac{\mathrm{d}^{m-1}}{\mathrm{d}s^{m-1}}\left[(s-p_k)^m X(s)\mathrm{e}^{st}\right]_{s=p_k}$$

Example: Find the inverse transformation $x(t)$ of $X(s)=\dfrac{1}{s^2+3s+2}$, $\mathrm{Re}(s)>-1$ using the residue method.

Solution: According to the residue method, the time domain expression of $X(s)$ can be obtained as:

$$x(t)=\begin{cases}\dfrac{1}{2\pi \mathrm{j}}\displaystyle\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty} X(s)\mathrm{e}^{st}\mathrm{d}s=\operatorname*{Res}_{s=-1}[X(s)\mathrm{e}^{st}]+\operatorname*{Res}_{s=-2}[X(s)\mathrm{e}^{st}] & t>0\\ 0 & t<0\end{cases}$$

$s_1=-1$ and $s_2=-2$ both $X(s)\mathrm{e}^{st}$ are single poles, and the corresponding residues are:

$$\operatorname*{Res}_{s=-1}[X(s)\mathrm{e}^{st}]=\frac{1}{s+2}\mathrm{e}^{st}\Big|_{s=-1}=\mathrm{e}^{-t}$$

$$\operatorname*{Res}_{s=-2}[X(s)\mathrm{e}^{st}]=\frac{1}{s+1}\mathrm{e}^{st}\Big|_{s=-2}=-\mathrm{e}^{-2t}$$

Therefore, $x(t)=(\mathrm{e}^{-t}-\mathrm{e}^{-2t})u(t)$

2. Partial fraction expansion method

※Partial fraction expansion

One-sided Laplace transforms of many signals into rational fraction forms:

$$X(s)=\frac{N(s)}{D(s)}=\frac{b_m s^m+b_{m-1}s^{m-1}+\cdots+b_1 s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1 s+a_0}$$

Question: How to solve the time domain representation corresponding to the above formula?

Example 1: Using the partial fraction expansion method to find the inverse transformation $x(t)$ of $X(s)$:

$$X(s)=\frac{2}{s^2+3s},\quad \mathrm{Re}(s)>0$$

Solution:

$X(s)$is a rational proper fraction with only a single root of the denominator polynomial:

$$X(s)=\frac{2}{s(s+3)}=\frac{k_1}{s}+\frac{k_2}{s+3}$$

Multiply both sides of the above formula by s to get:

$$sX(s)=\frac{2}{s+3}=k_1+\frac{sk_2}{s+3}$$

Let s=0, only the k_1 item on the right side of the above formula is not equal to zero, so

$$k_1=sX(s)\big|_{s=0}=\frac{2}{s+3}\Big|_{s=0}=\frac{2}{3}$$

Example 1: Using the partial fraction expansion method to find the inverse transformation $x(t)$ of $X(s)$:

$$X(s)=\frac{2}{s^2+3s},\quad \mathrm{Re}(s)>0$$

Solution: Can be obtained in the same way

$$k_2=(s+3)X(s)\big|_{s=-3}=\frac{2}{s}\Big|_{s=-3}=-\frac{2}{3}$$

Therefore

$$X(s)=\frac{2}{s(s+3)}=\frac{2/3}{s}+\frac{-2/3}{s+3}$$

Inverse Laplace transform of the above formula can be obtained

$$x(t)=\frac{2}{3}u(t)-\frac{2}{3}\mathrm{e}^{-3t}u(t)$$

Example 2: Using the partial fraction expansion method to find the inverse transformation $x(t)$ of $X(s)$:

$$X(s)=\frac{s-1}{s(s+1)^2},\quad \mathrm{Re}(s)>0$$

Solution: $X(s)$ is a rational proper fraction, and the denominator polynomial has 1 double root:

$$X(s)=\frac{k_1}{s}+\frac{k_2}{(s+1)^2}+\frac{k_3}{s+1} \qquad ①$$

$$k_1=sX(s)\big|_{s=0}=\frac{s-1}{(s+1)^2}\Big|_{s=0}=-1$$

Multiply both sides of formula ①by$(s+1)^2$ to get

$$(s+1)^2X(s)=\frac{k_1(s+1)^2}{s}+k_2+k_3(s+1) \qquad ②$$

Let $s=-1$, ②only the k_2 item on the right side of the formula is not equal to zero, so

$$k_2=(s+1)^2X(s)\big|_{s=-1}=\frac{s-1}{s}\Big|_{s=-1}=2$$

Example 2: Using the partial fraction expansion method to find the inverse transformation $x(t)$ of $X(s)$:

$$X(s)=\frac{s-1}{s(s+1)^2},\quad \mathrm{Re}(s)>0$$

Solution:

$$(s+1)^2X(s)=\frac{k_1(s+1)^2}{s}+k_2+k_3(s+1) \qquad ②$$

Calculate the first derivative of formula ② and then set $s=-1$ to get:

$$k_3=\frac{\mathrm{d}}{\mathrm{d}s}\left[(s+1)^2X(s)\right]\Big|_{s=-1}=\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{s-1}{s}\right)\Big|_{s=-1}=1$$

Therefore,

$$X(s)=\frac{-1}{s}+\frac{2}{(s+1)^2}+\frac{1}{s+1}$$

$$x(t)=\mathscr{L}^{-1}[X(s)]=(-1+2t\mathrm{e}^{-t}+\mathrm{e}^{-t})u(t)$$

Example 3: Use the partial fraction expansion method to find the inverse transformation $x(t)$ of the following $X(s)$:

$$X(s)=\frac{s^3-s^2-12s+2}{s^2+3s},\quad \mathrm{Re}(s)>0$$

Solution:

$X(s)$ is a rational false fraction, which can be transformed into a rational true fraction:

$$X(s)=s-4+\frac{2}{s^2+3s}$$

$$\delta'(t)\xleftrightarrow{\mathscr{L}}s$$

$$\delta(t)\xleftrightarrow{\mathscr{L}}1$$

Use Example 1 to calculate the result

$$x(t)=\delta'(t)-4\delta(t)+\frac{2}{3}u(t)-\frac{2}{3}\mathrm{e}^{-3t}u(t)$$

2. Partial fraction expansion method

Summary $$X(s)=\frac{N(s)}{D(s)}=\frac{b_m s^m+b_{m-1}s^{m-1}+\cdots+b_1 s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1 s+a_0}$$

1) If $X(s)$ is a rational true fraction ($m < n$), and the denominator polynomial has only a single root, then

$$X(s)=\frac{N(s)}{D(s)}=\frac{N(s)}{(s-p_1)(s-p_2)\cdots(s-p_n)}$$

$$=\frac{k_1}{s-p_1}+\frac{k_2}{s-p_2}+\cdots+\frac{k_n}{s-p_n}$$

Then $$k_i=(s-p_i)X(s)\Big|_{s=p_i} \qquad i=1,2,\cdots,n$$

$$x(t)=(k_1 e^{p_1 t}+k_2 e^{p_2 t}+\cdots+k_n e^{p_n t})u(t)$$

2. Partial fraction expansion method

$$X(s)=\frac{N(s)}{D(s)}=\frac{b_m s^m+b_{m-1}s^{m-1}+\cdots+b_1 s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1 s+a_0}$$

2) If $X(s)$ is a rational true fraction($m < n$), and the denominator polynomial has multiple roots, then

$$X(s)=\frac{N(s)}{D(s)}=\frac{N(s)}{(s-p_1)^r(s-p_{r+1})\cdots(s-p_n)}$$

$$=\boxed{\frac{k_1}{(s-p_1)^r}+\frac{k_2}{(s-p_1)^{r-1}}+\cdots+\frac{k_r}{s-p_1}}+\frac{k_{r+1}}{s-p_{r+1}}+\cdots+\frac{k_n}{s-p_n}$$

$$k_i=\frac{1}{(i-1)!}\frac{\mathrm{d}^{i-1}}{\mathrm{d}s^{i-1}}[(s-p_1)^r X(s)]\quad i=1,2,\cdots,r$$

$$x(t)=[\sum_{i=1}^{r}\frac{k_i}{(r-i)!}t^{r-i}\mathrm{e}^{p_1 t}]u(t)+\sum_{i=r+1}^{n}k_i\mathrm{e}^{p_i t}u(t)$$

2. Partial fraction expansion method

$$X(s)=\frac{N(s)}{D(s)}=\frac{b_m s^m+b_{m-1}s^{m-1}+\cdots+b_1 s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1 s+a_0}$$

3) $X(s)$is a rational pseudo-fraction($m\geq n$), then

$$X(s)=\frac{N(s)}{D(s)}=B_0+B_1 s+\cdots+B_{m-n}s^{m-n}+\frac{N_1(s)}{D(s)}$$

$$B_0 \xleftrightarrow{\mathscr{L}} B_0\delta(t)$$

$$B_1 s \xleftrightarrow{\mathscr{L}} B_1\delta'(t)$$

$$B_{m-n}s^{m-n} \xleftrightarrow{\mathscr{L}} B_{m-n}\delta^{(m-n)}(t)$$

$\frac{N_1(s)}{D(s)}$ is a true fraction, according to the situation is expanded according to 1) or 2).

Bilateral Laplace transform and inverse transform

- Definition of Bilateral Laplace Transform
- Properties of Bilateral Laplace Transform
- Bilateral inverse Laplace transform

Definition of Bilateral Laplace Transform

Bilateral Laplace transform:

Forward Laplace transform: $$X(s)=\int_{-\infty}^{\infty} x(t)\mathrm{e}^{-st}\mathrm{d}t$$

Inverse Laplace transform: $$x(t)=\frac{1}{2\pi \mathrm{j}}\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty} X(s)\mathrm{e}^{st}\mathrm{d}s$$

Definition of Bilateral Laplace Transform

$$X(s)=\int_{-\infty}^{\infty} x(t)\mathrm{e}^{-st}\mathrm{d}t$$

If the bilateral Laplace transform of $x(t)$ exists, the above integral needs to converge.

Therefore, the necessary and sufficient conditions for the existence of bilateral Laplace transform are:

$$\int_{-\infty}^{\infty} |x(t)\mathrm{e}^{-st}|\mathrm{d}t=\int_{-\infty}^{\infty} |x(t)\mathrm{e}^{-\sigma t}|\mathrm{d}t=C, \qquad \sigma=\mathrm{Re}(s)$$

■The value range of σ where the above formula holds is called the convergence region of Laplace transform,

Abbreviated as ROC(Region Of Convergence).

Definition of Bilateral Laplace Transform

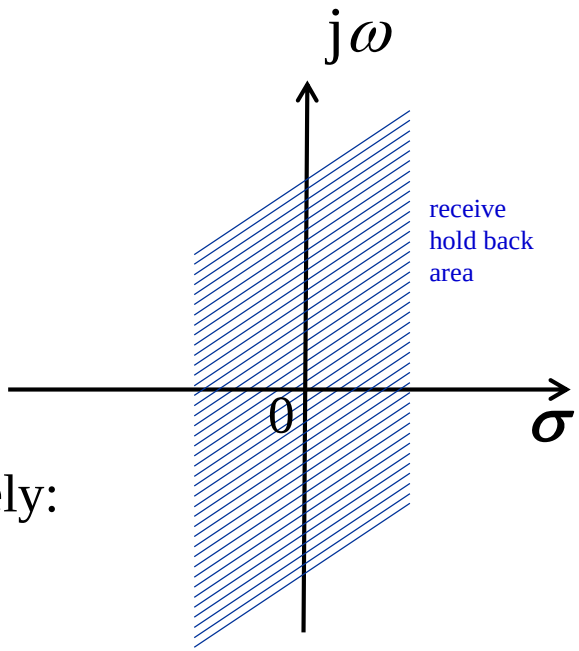
(1) Limited length signal

Example: Try to find the bilateral Laplace transform of continuous signal $u(t+2)-u(t-2)$ and its convergence region.

Solution: $X(s)=\int_{-\infty}^{\infty}[u(t+2)-u(t-2)]\mathrm{e}^{-st}\mathrm{d}t$

$$=\int_{-2}^{2}\mathrm{e}^{-st}\mathrm{d}t=-\frac{1}{s}\mathrm{e}^{-st}\Big|_{-2}^{2}$$

$$=\frac{1}{s}(\mathrm{e}^{2s}-\mathrm{e}^{-2s})$$



The region of convergence is the *s*-full plane, namely:

$$\mathrm{Re}(s)>-\infty$$

Definition of Bilateral Laplace Transform

(2) Right signal

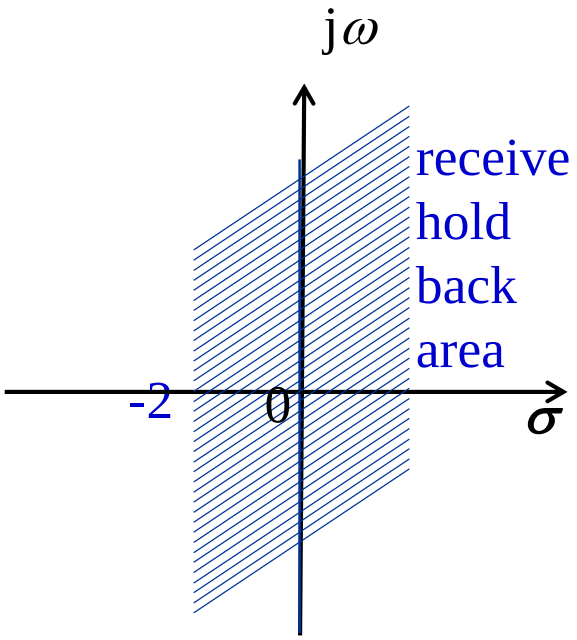
Example: Try to find the bilateral Laplace transform of continuous signal $e^{-2t}u(t)$ and its convergence region.

Solution: $X(s)=\int_{-\infty}^{\infty} e^{-2t}u(t)e^{-st}dt$

$$=\int_{0}^{\infty} e^{-2t}\cdot e^{-st}dt$$

$$=-\frac{1}{s+2}e^{-(s+2)t}\Big|_{0}^{\infty}$$

$$=\frac{1}{s+2}$$



Domain of convergence: $\mathrm{Re}(s)>-2$

Definition of Bilateral Laplace Transform

(3) Left signal

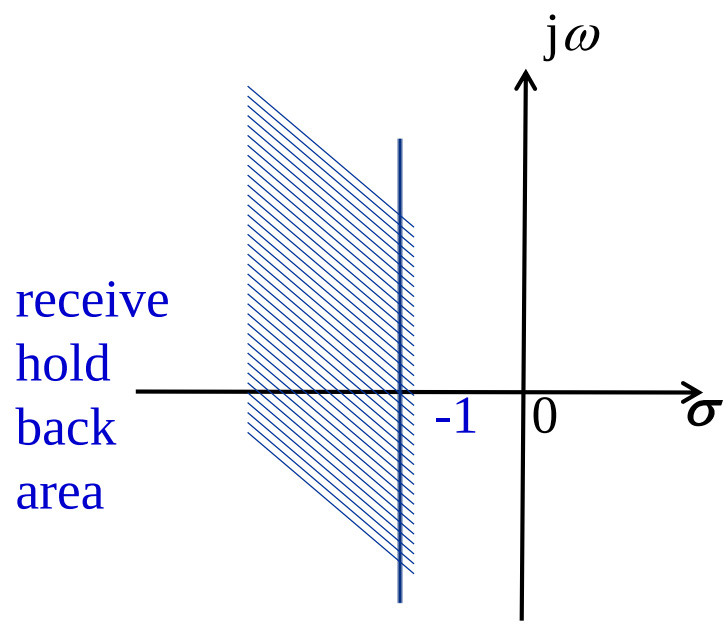
Example: Try to find the bilateral Laplace transform of continuous signal $\mathrm{e}^{-t}u(-t)$ and its convergence region.

Solution: $$X(s)=\int_{-\infty}^{\infty}\mathrm{e}^{-t}u(-t)\cdot\mathrm{e}^{-st}\mathrm{d}t$$

$$=\int_{-\infty}^{0}\mathrm{e}^{-t}\cdot\mathrm{e}^{-st}\mathrm{d}t$$

$$=-\frac{1}{s+1}\mathrm{e}^{-(s+1)t}\Big|_{-\infty}^{0}$$

$$=\frac{-1}{s+1}$$



The region of convergence is $\mathrm{Re}(s)<-1$

Definition of Bilateral Laplace Transform

(4) Bilateral signal

Example: Try to find the bilateral Laplace transform of continuous signal $e^{-t}u(-t)+e^{-2t}u(t)$ and its convergence region.

Solution: $X(s)=\int_{-\infty}^{0} e^{-t}u(-t)e^{-st}dt+\int_{0}^{\infty} e^{-2t}u(t)e^{-st}dt$

$$=\frac{-1}{s+1}+\frac{1}{s+2}$$

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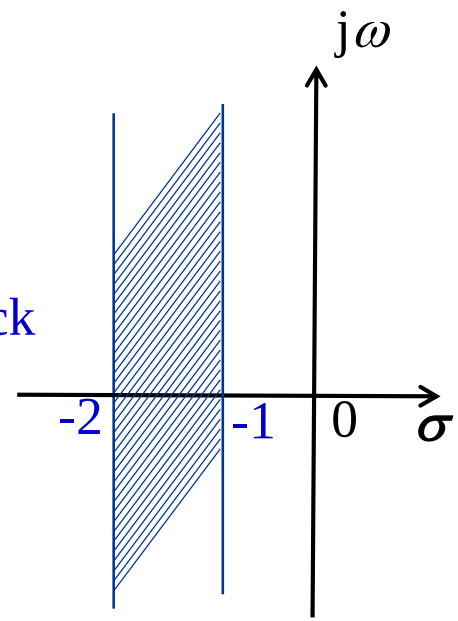

The region of convergence is: $-2<\mathrm{Re}(s)<-1$

Properties of Bilateral Laplace Transform

► time-shifting characteristics

If $x(t) \xleftrightarrow{\mathscr{L}} X(s)$, $\mathbf{R}_x = \{s; \sigma_1 < \mathrm{Re}\{s\} < \sigma_2\}$

then $x(t-t_0) \xleftrightarrow{\mathscr{L}} \mathrm{e}^{-st_0} X(s)$, $\mathbf{R}_x$

Properties of Bilateral Laplace Transform

- Differential characteristics

If $x(t) \xleftrightarrow{\mathscr{L}} X(s), \quad \mathbf{R}_x = \{s; \sigma_1 < \mathrm{Re}\{s\} < \sigma_2\}$

Then $\dfrac{\mathrm{d}x(t)}{\mathrm{d}t} \xleftrightarrow{\mathscr{L}} sX(s), \quad \mathbf{R}_x$

Properties of Bilateral Laplace Transform

► Integral characteristics

If $x(t) \xleftrightarrow{\mathscr{L}} X(s), \quad \mathbf{R}_x = \{s; \sigma_1 < \mathrm{Re}\{s\} < \sigma_2\}$

Then $\int_{-\infty}^{t} x(\tau)\mathrm{d}\tau \xleftrightarrow{\mathscr{L}} \dfrac{X(s)}{s}$

$$\boldsymbol{R}_x \cap \{\mathrm{Re}\{s\} > 0\}$$

Bilateral inverse Laplace transform

$$x(t)=\frac{1}{2\pi \mathrm{j}}\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty}X(s)\mathrm{e}^{st}\mathrm{d}s$$

※ Residue method

The residue method is more complicated to calculate, but has a wider scope of application.

※ Partial fraction expansion

The partial fraction method is relatively simple to solve, but it is generally only applicable to rational fractions.

Bilateral inverse Laplace transform

※Partial fraction expansion

$$e^{\lambda t}u(t) \xleftrightarrow{\mathscr{L}} \frac{1}{s-\lambda}, \quad \mathrm{Re}(s) > \lambda$$

$$te^{\lambda t}u(t) \xleftrightarrow{\mathscr{L}} \frac{1}{(s-\lambda)^2}, \quad \mathrm{Re}(s) > \lambda$$

$$-e^{\lambda t}u(-t) \xleftrightarrow{\mathscr{L}} \frac{1}{s-\lambda}, \quad \mathrm{Re}(s) < \lambda$$

$$-te^{\lambda t}u(-t) \xleftrightarrow{\mathscr{L}} \frac{1}{(s-\lambda)^2}, \quad \mathrm{Re}(s) < \lambda$$

Example: According to the convergence domain of $X(s)$, solve the corresponding time domain signal $x(t)$ respectively.

$$X(s)=\frac{2s+1}{(s+2)(s+3)}$$

Solution:

$$X(s)=\frac{-3}{s+2}+\frac{5}{s+3}$$

(1) $\mathrm{Re}(s)>-2$

$$x(t)=-3\mathrm{e}^{-2t}u(t)+5\mathrm{e}^{-3t}u(t)$$

(2) $-3<\mathrm{Re}(s)<-2$

$$x(t)=3\mathrm{e}^{-2t}u(-t)+5\mathrm{e}^{-3t}u(t)$$

(3) $\mathrm{Re}(s)<-3$

$$x(t)=3\mathrm{e}^{-2t}u(-t)-5\mathrm{e}^{-3t}u(-t)$$

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!